

Vaping: Curriculum Information for Administrators and Teachers

September 2019

CURRENT SITUATION

- The federal governments in the US and Canada are investigating a number of vaping-related illnesses and deaths. The US began documenting cases in April 2019. The first Canadian case was documented in September 2019. The majority of cases are among youth and young adults.
- Canada and the provinces and territories, including Nova Scotia, are implementing surveillance systems to ensure cases are identified when seen by healthcare providers.
- As a relatively new product, the long-term harms of e-cigarettes are not yet known. For the past several years, health practitioners have thought e-cigarettes were less likely to have long-term health effects compared to tobacco. However, emerging research on the associated health risks are reinforcing the fact that there are many unknowns regarding the safety of vaping products in both the short- and long-term.

BACKGROUND

Tobacco rates among youth have declined over the past twenty years. This has been due to a variety of policies and legislation at the provincial and federal level. When the parents and grandparents of today's youth grew up, tobacco use was not only normalized in everyday life, but also on television, and in the movies. As more regulations were put on tobacco in the last thirty or so years, the social norms around tobacco use have contributed to its decline in use among youth and adults.

However, there is concern that this may change as we see increasing numbers of youth vaping in Canada, and in Nova Scotia.

Vaping was introduced into the Canadian market around 2007, marketed as a quit aid for adult smokers. It lacked the same regulation as tobacco products, which has opened the door for industry to bring in high nicotine vaping products alongside advertising (via some currently unregulated media channels) promoting misconceptions that vaping products are less harmful than cigarette smoking.

Vaping devices come in sleek high-tech designs that appeal to youth because of their novel, cool and harmless appearance. Vaping products come in a variety of attractive flavours and the majority contain nicotine. Furthermore, the use of these products is being widely shared via social media. Most concerning are the new types of vaping products on the market, like pods and salts, that contain very high levels of nicotine. A single nicotine pod can expose a user to the same amount of nicotine as an entire pack of cigarettes.

It is important to keep in mind that while most youth are not vaping, we are seeing an alarming number of non-smoking youth vaping in Canada. This trend is very troubling, especially because of what we know about nicotine.

Nicotine is highly addictive and can have harmful impacts on the brain, affecting memory and concentration in everyone and brain development in youth and young adults. It alters parts of the brain that control attention, learning, mood, and impulse control. Early exposure to nicotine in adolescence may increase the severity of future dependence to nicotine and tobacco.

While Health Canada has been working over the course of the year to tighten regulations on vaping/vaping products, youth in Canada/Nova Scotia are seeing images and promotion of vaping and vaping products via social media settings, and in some cases among their peer group.

What is vaping?

Vaping involves using a device called an e-cigarette to aerosolize and inhale a liquid solution.

All e-cigarettes have common components that include a battery, heating coil, atomizer that transforms the vape liquid into an aerosol, a cartridge that contains the vape liquid, and mouthpiece.

While the activity is usually referred to as vaping, the e-cigarette device used to vape has many common names including mods, vape pens, e-cigs, mod pods, JUULs, vape sticks, as well as other names that continue to emerge.

IMPACT ON THE HEALTH EDUCATION CURRICULUM

For youth and young adults, vaping carries increased health harms because their brains are still developing. Given this context, schools have an opportunity to help students be well-informed about the risks of vaping. At the same time, teachers can continue to reinforce curriculum components that address substance misuse, including social-emotional learning, media literacy, healthy relationships, and mental and emotional health.

Substance misuse, also known as drug education, is one of six priority health areas within the grades Primary to 9 health education curricula. Lessons addressing substance misuse vary at different grade levels, as well as at the different stages of cognitive, social, and emotional development. Drug education in Nova Scotia takes these differences into account so that it is age and developmentally appropriate, focuses on risk and protective factors, and enables teachers and administrators to address local substance use patterns.

The following sections highlight how and where substance misuse and its related health concepts are addressed in the health education curriculum between grades Primary –9 and how health education can be woven into high school curriculum.

Grades Primary to 3

Nova Scotia's health education curriculum in the early grades is critical to setting the stage for healthy development as students move through childhood and into adolescence. Many of the concepts introduced to students in grades P–3 are protective factors against substance misuse and include

- discussing healthy ways to cope with change
- naming and discussing a growing variety of emotions and emotional responses to daily life
- investigating the qualities of healthy relationships with friends and family
- investigating safe and unsafe scenarios at home, school, and in the community
- discussing the importance of having safe grownups/adults in their lives and how to ask a safe grown-up/adult for help
- what to do if you find unsafe substances at home, on the playground, or elsewhere

The Canadian Centre for Child Protection's *Kids in the Know: A Personal Safety Program* remains a key resource for classroom teachers. Its lessons focus on

- dealing with emotions
- staying safe in various environments
- practicing assertiveness and refusal skills
- identifying a safe adult in their lives; how to ask for help
- what makes a good friend

This focus helps students to build transferrable, health-enhancing skills that can transition from childhood into adolescence. The *Kids in the Know: A Personal Safety Program* was distributed to schools with grades P–9 and should be found in schools with these grade levels. Grade level supplements are available through the School Book Bureau. The website has more information in English: www.kidsintheknow.ca/app/en/ and in French: www.kidsintheknow.ca/app/fr/index.

Grades 4 to 6

In grades 4–6, friendships can become more challenging, physical and emotional development is beginning to change, and children feel the impact of stressors. Students at this grade level begin to have fuller lives and can be impacted by pressures and messages from their social environment and various forms of media (e.g., marketing, advertising, social media).

The health education curriculum continues to focus on health concepts that are protective factors against substance misuse with more complexity than in grades P–3, such as

- ways to enhance mental health
- recognizing difficult emotions, identifying their purpose, and coping with stressors in positive ways
- understanding the characteristics of healthy and unhealthy friendships
- recognizing techniques of marketing and advertising of unhealthy substances including foods, beverages, alcohol, tobacco products, and cannabis
- recognizing benefits and harms of new technologies, including the internet and social media sites

The recommendations from the document, *16 Best Practices in School-Based Drug Education for Grades 7–9* (Nova Scotia Department of Health and Wellness 2007) was used to inform the outcomes of the Nova Scotia Public School Health Education Curriculum. Particular attention was paid to the document's recommendations to not focus on substances for which there is not yet a pattern of use in a population, and to not teach units and lessons focusing on a single drug because research shows these are only effective after students reach 14 years of age.

www.novascotia.ca/dhw/addictions/documents/16-Best-Practices-in-School-Based-Drug-Education-for-Grades-7-9.pdf.

Lessons that address the misuse of substances like alcohol, cannabis, tobacco products (including vaping), and prescription drugs are most effective when they are participatory, interactive, and focused on building health skills such as decision-making, critical thinking, negotiation, assertiveness, and refusal skills. Students learn these skills best when teachers use authentic scenarios that they will face during their teenage years. Effective drug education is not simply the acquisition of information on drug types and their effects.

EECD recently approved a set of resources from Pearson Canada integrating English language arts with health education. This is a series of short books by Canadian authors including a teacher resource for grades 4–8. The series provides an awareness of mental health issues and fosters conversation on a wide variety of topics related to emotional well-being. *Well Aware* (complete classroom package for each grade 4–8) is newly listed on the Authorized Learning Resource list.

Grades 7 to 9

Effective delivery of health education in these grades provides topic coverage through interactive learning experiences that are conversation-based and non-judgmental. Critical life skills are practiced, equipping students with the ability to apply these skills in real life risk scenarios. They begin to analyze their social influences to use and practice ways to refuse while learning the risks of substances that are prevalent among youth in their communities, which are primarily alcohol and cannabis.

A Question of Influence: A Drug Education Resource for Grades 7-9 (2008) contains interactive lessons and background information for teachers and students. It is available in English and French.

EECD is working with the Nova Scotia Health Authority to update *A Question of Influence* and post its content online. It will include lessons created to address the emerging issue of youth vaping, and strengthen attention to prescription drug misuse, a concern in Nova Scotia and across Canada. These lessons and their appendices address substance misuse and are very much applicable to cannabis use, as well as alcohol and other drugs, and can be adapted to address vaping.

Lessons on various substance use (including two versions of a lesson on vaping) and other resources, such as a PDF to the drug education curriculum supplement *A Question of Influence* and links to websites related to vaping are accessible to all staff via Google Drive with GNSPES/SEPNE login credentials. To access the documents

1. Copy and paste the following into the search bar in Google Drive: **source:domain owner:healthed@gnsps.ca type:folder**
2. There will be a folder titled "Drug Education".
3. A copy of the folder can be added to teachers' own Google Drive directories.

EECD will continue to identify high-quality resources and lessons to include in the health education curriculum for grades 7–9 on vaping as they developed and/or identified and make them available via Google Drive.

Grades 10 to 12

While health education is not offered as a discrete subject area in grades 10–12, there are outcomes in multiple subject areas, as well as resources that address vaping and other substance misuse for these grade levels. Youth health centres, SchoolsPlus, guidance counsellors, and school staff play an important role in health promotion in the high school environment.

Know Before You Go is an online resource in French and English with a section on drug misuse. It is available as a downloadable PDF from TeenMentalHealth.org: www.teenmentalhealth.org/product/know-before-you-go/.

It is also available as an interactive online resource through EECD's website in English: knowbeforeyg.ednet.ns.ca/ and in French: knowbeforeyg.ednet.ns.ca/fr/.

Biology 12 offers an opportunity for teachers to discuss key messages around substance misuse in the unit that explores influences on the nervous systems and addresses the health education outcome of, "Analyze how and why technologies and drugs developed and improved over time can affect homeostasis (115-5, 317-7)."

Guest Presentations

Schools are often approached by various individuals and organizations wishing to provide information on the impacts of drugs and such presentations that address youth vaping have and will continue to emerge. Guest presentations are an important mechanism for making connections to organizations and partners in the broader community.

Nevertheless, advanced planning around the role of guest presentations in drug education will help avoid enabling or glamorizing drug behaviour with student populations.

Guests who are invited to deliver drug education sessions, need to be able to address curricular goals and work interactively with the students, rather than present an isolated session unconnected to the curriculum. We encourage school administrators to work through Youth Health Centre coordinators and/or Schools Plus staff to connect with colleagues in EECD and/or NSHA for recommendations on the value and efficacy of guest speakers that might wish to access youth through school presentations.

GENERAL INFORMATION

The Importance of Protective Factors

Many factors influence youth substance use. Effective drug education strategies can improve positive outcomes by focusing on reducing risk factors and increasing protective factors. Protective factors mitigate the impact of risk factors, such as peer, family, and community influences, as well as mental health issues.

Protective factors include

- . having healthy relationships with friends, family, and community
- . choosing positive ways to cope with and manage stress, such as physical activity, healthy eating, and getting enough sleep
- . knowing how and who to ask for help in maintaining mental health and/or for problematic substance use

Helping students build protective factors is more than drawing attention to risk factors or seeking to inoculate against risk of substance use/misuse. Schools already promote and enhance protective factors through

- . the effective use of instructional time in health education classes
- . extracurricular activities
- . intramurals
- . physical education classes
- . physically active time
- . healthy eating programs
- . SchoolsPlus programing
- . other health promoting additions that provide opportunities for students to practice their skills in authentic learning experiences

These opportunities build the capacity of children and youth to adapt to changing circumstances and deal with past, present, and future challenges in healthy ways. The focus of drug education is youth resilience in a world where drugs are easily obtained, and where their use is often promoted.

A key element in nurturing resilience against drug use during students' childhood and adolescent years is enhancing their sense of belonging and connectedness. Well-being involves embodied experiences of connectedness with school, community, peers, and caring adults. Students need to have individuals in their lives who are willing and equipped to have open and ongoing conversations with them about the challenges they may face and the importance of making decisions that will promote both their own health and the well-being and health of others. Adolescence is a time when they will encounter new and different pressures and challenges than they may have previously experienced.

A new resource, recently published by Nelson Canada called *The Third Path: A Relationship Approach to Student Well-being and Achievement* (2018) by David Tranter, Lori Carson, and Tom Boland offers a practical and promising approach that addresses eight conditions that nurture well-being and achievement. It is a practical resource that everyone in the school community can use to build positive relationships with all students. It is available from the Nova Scotia School Book Bureau in English.

RESOURCES

Consistent Messaging on Youth Vaping

Children and youth benefit when the adults at home, school, and in their communities share consistent messages. Health Canada has a webpage with helpful information for educators and school staff to become familiar with what we know about vaping currently — devices used, its harms, what families/parents need to know, vaping and youth, current national statistics, and also has some youtube video clips. The following is a list of links to reliable Canadian sources that may be helpful for

administrators, teachers, counsellors, and families when talking with students about the harms associated with vaping:

From Health Canada:

<https://www.canada.ca/en/health-canada/services/smoking-tobacco/vaping/canada.html>

From the Nova Scotia Lung Association:

<https://ns.lung.ca/lung-health/vaping-and-nova-scotia-youth>

Though a PEI website, this page has very good information, overview, and an infographic on vaping.

<http://www.peitobaccoreduction.com/vape/>

Youth Engagement

The Pan-Canadian Joint Consortium on School Health (JCSH) has recently updated and revised their guidelines and background information on youth engagement. Its youth engagement tool is intended to raise awareness of the importance of youth engagement in building school health. It can assist school communities with practical steps on how to build the conditions for school health. It is available in English: www.jcsh-cces.ca/index.php/resources/latest-website-additions/143-youth-engagement-toolkit and in French: www.jcsh-cces.ca/index.php/ressources/engagement-des-jeunes.